NI TB Genomic Surveillance

Outbreak Investigation Report

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REPORT STATUS: DEVELOPMENT / INTERNAL DRAFT ONLY - Required reproducibility metadata is incomplete; external circulation is blocked until mandatory fields are populated.

Executive Action Summary

Use this page first. Items marked model-only or exploratory require genomic validation and epidemiological corroboration before operational action.

Domain	Current finding	Risk	Immediate action
Circulation status	Development / Internal Draft Only	High	Complete reproducibility metadata before external governance circulation
Sequencing coverage	83.3%	Low	Maintain / widen only if capacity permits
QC pass rate	60.0%	High	Review failed and unreported samples
Contamination	1	Moderate/high	Repeat sequencing / exclude from cluster assignment
Open clusters	3	High	Confirm investigation leads
Model-prioritised links >0.70	18	Moderate	Validate against pairwise SNP and epidemiology
MDR/RR signals	13	High	Confirm with phenotypic DST
Model reliability	Exploratory	High uncertainty	Do not overinterpret directionality

Top Actions Due Now

Priority	Action	Owner	Due
1	Repeat sequencing / QC review	Laboratory	48 h
2	Validate drug-resistance pipeline mapping	Bioinformatics / Microbiology	Immediate
3	Confirm phenotypic DST	TB Microbiology / MDT	Immediate
4	Review open genomic clusters	TB MDT / PHA	Next MDT
5	Do not escalate model-only links to field	HPT / TB Nurses	Ongoing

Table 1. Operational worklist for immediate MDT actioning.

Reasons: 1 — QC: 8 low-coverage fails, 4 QC not-reported, 1 contamination flags. 2 — Resistance: Gene-drug combinations are preliminary; verify against WHO catalogue. 3 — DST: MDR/RR genomic signals require culture-based confirmation before clinical action. 4 — Clusters: 3 clusters remain open for epidemiological investigation. 5 — Model links: No pairwise SNP ≤12 direct-transmission support in this extract.

MDT Governance Summary

Priority area	Current signal	Required MDT action	Owner	When
Circulation readiness	Development / Internal Draft Only	Complete mandatory reproducibility metadata before external circulation	Pipeline + Governance	Before circulation
Model reliability	Exploratory	Treat directionality as exploratory; diagnostics unavailable	Bioinformatics + MDT	Current cycle
Open clusters	3 total / 2 priority >10	MDT review and epi data completion for all open clusters	MDT + Field team	Next MDT
Discordant model links	see Appendix B	Pairwise SNP + epi adjudication pathway	MDT	Next MDT
Geography completeness	UK-only extract	Re-ingest with HSC Trust / PHA locality / council area	Data management	Next ingest

Full MDT action sheet with final discordant-pair counts is in **Appendix C**.

Contents

Appendices A–D are in landscape format after the main report. Flag codes used in pair tables: SNP-linked = SNP ≤12 supported; D1: SNP>12 = genomically discordant; Model-only = no pairwise SNP; QC-unresolved = QC failed/not reported.

About This Report

This report is produced by the Northern Ireland TB Genomic Surveillance platform using whole-genome sequencing (WGS) data and epidemiological case records. It is intended to support TB programme staff and public health investigators by providing genomic evidence for transmission clusters, drug-resistance profiles, and programme performance metrics. **This is a decision-support tool only — all findings must be reviewed and acted on by a qualified clinician or public health professional. No automated decisions are made.**

Genomic terminology: For definitions of WGS, SNP, cluster, outbreaker2, MCMC convergence, and other terms used in this report, see **Appendix D: TB Genomics Key Concepts** at the end of this document.

Case Summary

Total Cases	24
Clustered Cases	14
Unclustered Cases	10
Open Clusters	3

Table 2. Programme case count summary. Clustered cases are those linked by genomic similarity to at least one other case. Unclustered (singleton) cases may represent imported strains, sporadic transmission, or reactivation of latent disease. Open clusters are active genomic transmission clusters with ongoing epidemiological investigation.

Automated Interpretation Flags

- Open clusters requiring investigation: 3.
- Model-prioritised transmission hypotheses present; validate before field escalation.

Analysis Summary

Posterior Samples	1500
MCMC Iterations	2000
Burn-in	500
Mean Log-Likelihood	-18376.15
Log-Likelihood SD	3618.29

Table 3. outbreaker2 Bayesian MCMC analysis parameters. **Posterior Samples** is the number of accepted MCMC draws used to compute estimates — higher values give more stable posteriors. **Mean Log-Likelihood** reflects model fit; values closer to zero (less negative) indicate better fit. **Transmission Probability** is the average posterior probability that any given case-pair represents a direct transmission event. **Generation Time** is the modelled average interval (in days) between infection events in a transmission chain. **Convergence Diagnostic** near 1.0 confirms the MCMC chain has stabilised; values >1.1 indicate cautious interpretation is needed.

How to interpret outbreaker2 results: outbreaker2 reconstructs the most probable transmission tree using both genetic distance (SNPs) and timing (collection dates). Pairs with high posterior transmission probability are model-prioritised transmission hypotheses only. In this report, they should not be interpreted as direct transmission unless supported by pairwise SNP distance ≤12, QC pass status, and epidemiological corroboration. Lower probability pairs may still be linked within the same cluster but through one or more undetected intermediate cases.

Programme Surveillance KPIs (Last 12 Weeks)

Full surveillance KPI artifact unavailable at report generation time. However, high-level KPI summary is available from the Executive Action Summary: Sequencing coverage 83.3%, QC pass rate 60.0%, Contamination flags 1, Open clusters 3. For complete regional and temporal KPI trends, check the surveillance system directly.

Table 4. Regional sequencing representativeness. Regions with coverage <80% may introduce ascertainment bias — clusters in under-sequenced regions may be underdetected. Where persistent regional gaps exist, review laboratory submission pathways and specimen transport processes.

Weekly Surveillance Trends (12 Weeks)

Week	Eligible	Sequenced	Coverage %	QC Pass %
2026-02-23	0	0	None	None
2026-03-02	1	1	100.00	0.00
2026-03-09	1	1	100.00	100.00
2026-03-16	0	0	None	None
2026-03-23	0	0	None	None
2026-03-30	0	0	None	None
2026-04-06	0	0	None	None
2026-04-13	0	0	None	None
2026-04-20	0	0	None	None
2026-04-27	1	1	100.00	100.00
2026-05-04	0	0	None	None
2026-05-11	0	0	None	None

Table 5. Weekly sequencing coverage and QC pass rates over the last 12 weeks. A declining trend may indicate emerging laboratory issues. Weeks with zero eligible cases may reflect reporting lags rather than true absence of TB.

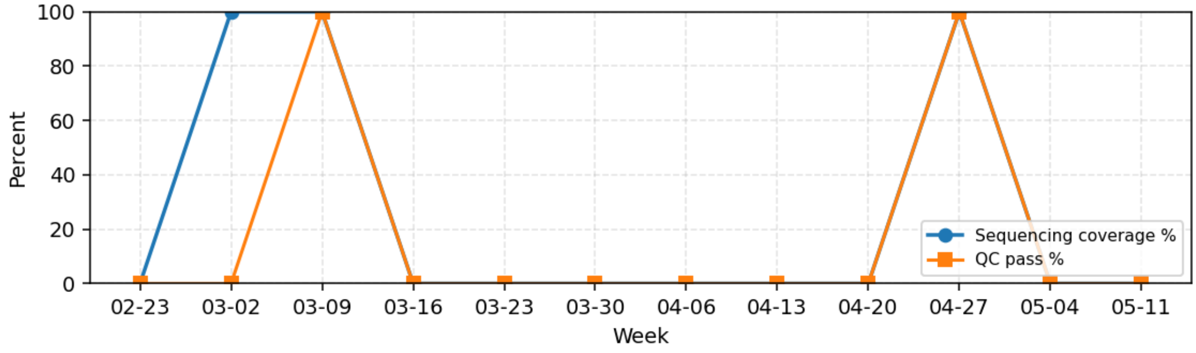


Figure 1. 12-week surveillance trend — sequencing coverage (%) and QC pass rate (%) by calendar week. Coverage is the proportion of eligible culture-confirmed TB cases that received WGS. Declining coverage weeks may reflect specimen submission delays, laboratory capacity issues, or data processing backlogs. QC pass rate below 90% in consecutive weeks warrants a laboratory review.

Cluster Action Prioritization

Cluster	Cases	Regions	Most Recent	Status	Priority
04e65e42	6	1	2026-05-02	open	21
39be6563	6	1	2026-01-25	open	18
604b7f43	2	1	2026-01-05	open	10

Table 6. Clusters ranked by investigation priority score. Score is composite: cluster size (×2), cross-region spread (×3), specimen recency within 14/30/60 days (×3/2/1), open investigation status (×3). Higher scores indicate clusters warranting prioritised MDT review and data-completeness follow-up. **Status 'open'** means an active review is ongoing or recommended.

Recommended action: For clusters with priority score >10 and status 'open', ensure MDT review and epidemiological data completion. Do not infer direct transmission or source-recipient direction from model output alone. Cross-region clusters should trigger coordination checks across NHS trust boundaries.

Lineage and Drug Resistance Validation

Interpreted Samples	24
Samples with Lineage	24
Lineage Coverage (%)	100.0
Samples with Resistance Calls	24
Resistance Coverage (%)	100.0
Any Resistance Signal	11
Rifampicin-Resistant (suspected)	11
Isoniazid-Resistant (suspected)	11
MDR (suspected)	11
FQ-Resistant (suspected)	11

Table 7. Lineage and drug-resistance epidemiology summary. Rows prioritize actionable burden indicators (coverage, resistance signal counts, and suspected RR/MDR/FQ resistance) to support triage and follow-up planning.

Top observed lineages: L4: 12, L2: 7, L3: 3, L1: 2

Secondary Transmission Engines

QC Filtering Status: The outbreaker2 network displayed below includes samples with reported QC status at run time. Samples with QC status 'fail' or 'not_reported' may be present in the graph. When using results operationally, assume all displayed transmission links are candidates pending post-hoc QC validation. Links involving QC-failed or not-reported samples should not be escalated to field investigation until repeat sequencing or QC review is complete.

No secondary engine validation artifact found.

Cross-Method Clustering Comparison

Sequence Assigned Cases	14
Outbreaker Assigned Cases	24
Overlap Cases	14
Pairwise Precision	0.341
Pairwise Recall	1.0
Pairwise Jaccard	0.341

Table 8. Agreement between SNP-threshold sequence clustering and outbreaker2 probabilistic clustering. **Precision**: of pairs grouped together by outbreaker2, the fraction also grouped by sequence clusters. **Recall**: of pairs grouped by sequence clusters, the fraction also grouped by outbreaker2. **Jaccard**: overall overlap index (0=no agreement, 1=perfect agreement). Discordant pairs — grouped by one method but not the other — may represent cases where temporal data (outbreaker2) overrides genomic distance alone, or where the SNP threshold is set differently.

Sequence Clustering Snapshot

Cluster Count	3

Case-Level Operational Actions

Detailed case classification (Table A1a) and recommended actions (Table A1b) are in Appendix A. Full data exported to exports/appendix_a_case_level_actions.csv.

Model-Prioritised Transmission Hypotheses

The outbreaker2 model infers transmission probabilities from SNP distance and sample collection dates. Posterior probability >0.70 indicates a plausible transmission event, but genomic validation is essential. Pairs are classified below by SNP support and QC status. **Do not escalate model-only or genomically discordant links to field investigation without genomic and epidemiological corroboration.**

■ **CRITICAL NOTE: No direct-transmission pairwise SNP links ≤12 are demonstrated in this report extract.** The outbreaker2 model has identified transmission hypotheses, but these currently lack clear genomic support within the recent-transmission threshold. All displayed model-prioritised links should be treated as hypotheses pending: (1) pairwise SNP analysis and validation, (2) QC review and repeat sequencing where needed, (3) epidemiological investigation to corroborate or refute the model's predictions. Do not escalate field investigations based on model probability alone.

Link category	Count
Pairwise SNP ≤5 and QC pass	0
Pairwise SNP 6–12 and QC pass	0
Pairwise SNP >12 and QC pass	13
SNP unavailable or QC unresolved	5
Total model-prioritised links ≥0.70	18

Pair	Post.	SNP	QC	Flag
fb419fc3->78e6e3e1	1.000	21	pass/pass	D1: SNP>12
78e6e3e1->f17d7f92	1.000	16	pass/pass	D1: SNP>12
a8d08022->b4218364	1.000	881	pass/pass	D1: SNP>12
f17d7f92->905ea01d	1.000	13	pass/pass	D1: SNP>12
a8d08022->e10f9dde	1.000	17	pass/pass	D1: SNP>12
e10f9dde->6f306409	1.000	18	pass/pass	D1: SNP>12
e10f9dde->631bcffb	1.000	16	pass/pass	D1: SNP>12

Table 9. Genomically discordant model predictions (posterior >0.70 but pairwise SNP >12). SNP distance does not support direct recent transmission; likely reflects extended genetic relatedness or model misspecification.

Pair	Post.	SNP	QC	Flag
89f900c2->3a23165b	1.000	n/a	not_reported/pass	QC-unresolved

Pair	Post.	SNP	QC	Flag
e0b496a8->24cc59c8	1.000	n/a	not_reported/fail	QC-unresolved
89f900c2->e0b496a8	1.000	n/a	not_reported/not_reported	QC-unresolved
24cc59c8->740131f2	1.000	n/a	fail/not_reported	QC-unresolved
3a23165b->ce1cd7ce	1.000	16	pass/fail	QC-unresolved
fb419fc3->b36fc4ca	1.000	28	pass/fail	QC-unresolved
b6530faa->1a574171	1.000	22	pass/fail	QC-unresolved
905ea01d->0ab71d84	1.000	15	pass/fail	QC-unresolved
716130a0->e5ffea1d	1.000	864	fail/fail	QC-unresolved
e0b496a8->b1548d7e	0.994	n/a	not_reported/pass	QC-unresolved
e10f9dde->9ab6c0d9	0.880	883	pass/fail	QC-unresolved

Table 10. Model-prioritised pairs involving QC-failed or not-reported samples. Hold pending repeat sequencing or QC review.

Current Outbreak Interpretation

Current interpretation: This report identifies three open genomic clusters and multiple outbreaker2 model-prioritised transmission hypotheses. However, no pairwise SNP links ≤12 are demonstrated in this extract, several links involve QC-failed or QC-not-reported samples, and model diagnostics remain exploratory. The immediate priorities are repeat sequencing/QC review, validation of resistance calls, phenotypic DST confirmation, and epidemiological corroboration before field escalation.

Counts used in this report

Three related but distinct counts appear in this report. They refer to different filtered views of the same model output.

Metric	Count	Definition
Model-prioritised links ≥0.70	18	All outbreaker2 edges with posterior probability ≥0.70 across full run
All discordant outbreaker2 links reviewed	see below	All model-linked pairs reviewed for adjudication, including pairs below the ≥0.70 threshold. This explains why Appendix B may include posterior values below 0.70 (e.g. 0.601 or 0.483).
Displayed/plotted network links	18	Links shown in network JSON snapshot (may be filtered for display)

Discordant model links: 23 total. With pairwise SNP >12: 17. With pairwise SNP unavailable: 6. Top five highest-priority discordant examples are shown below; the full table is provided in Appendix B.

Case Pair	Pairwise SNP	Posterior	Interpretation
0ab71d84-905ea01d	15	1.000	Temporal support without pairwise SNP support; assess missing intermediates/importation
3a23165b-89f900c2	n/a	1.000	Temporal support without pairwise SNP support; assess missing intermediates/importation
716130a0-e5ffea1d	864	1.000	Temporal support without pairwise SNP support; assess missing intermediates/importation
1a574171-b6530faa	22	1.000	Temporal support without pairwise SNP support; assess missing intermediates/importation
a8d08022-e10f9dde	17	1.000	Temporal support without pairwise SNP support; assess missing intermediates/importation

Table 11. Top discordant model-pair examples for MDT review.

Pairwise SNP Distance Summary

Pairwise SNP distances are the primary genomic evidence for or against direct recent transmission. The table below summarises all model-prioritised case pairs by SNP distance category. ≤12 SNPs is the operational threshold for probable recent transmission; >12 SNPs makes direct transmission unlikely; unavailable SNP (QC-failed or no consensus sequence) requires repeat sequencing before inference.

Table 12. Pairwise SNP distance for all model-prioritised case pairs. Pairs with SNP ≤12 and QC pass are primary candidates for direct transmission investigation. Pairs with SNP >12 or unavailable require genomic and epidemiological review before field action. Epi link column indicates whether epidemiological corroboration is available (same cluster, contact-traced, or unknown).

SNP distance category	Pair count	Operational implication
0–5 SNPs (direct)	0	Immediate: probable direct transmission — contact trace; confirm epi link
6–12 SNPs (probable)	0	Priority: probable cluster; review shared setting and exposures
13–25 SNPs (possible shared source)	9	Review: possible shared source/reactivation; epi adjudication required
>25 SNPs (unlikely direct)	4	Low priority: unlikely direct recent transmission; monitor only
SNP unavailable	5	Hold: repeat sequencing or QC resolution required before inference

QC Failure Drill-Down

QC category	Count	Meaning	Action
Fail: low coverage	8	Genome coverage below threshold	Repeat sequencing if culture is available
Fail: contamination	1	Mixed signal suspected	Exclude from cluster assignment pending repeat
Not reported	4	QC metadata missing	Do not use for operational inference until resolved
Pass	12	Acceptable for interpretation	Use with standard caveats
Excluded from operational interpretation	12	QC issue, contamination, or unresolved QC metadata	Hold from model-driven actions until resolved

Table 13. QC status summary and operational meaning. The summary distinguishes low coverage, contamination, not-reported metadata, and pass states so that missing data are not treated as failure by default.

Category	Operational meaning	Immediate use
Fail: low coverage	Genome below the interpretive threshold	Repeat sequencing if possible; do not use for directionality
Fail: contamination	Mixed or contaminated signal	Exclude from cluster assignment pending repeat
Not reported	QC metadata absent	Treat as unresolved and exclude from operational inference
Pass	Meets QC threshold	Use with standard caveats

Table 14. QC interpretation guide for operational staff.

Sample	QC	Coverage	Depth	Contam	Mixed-Lineage	Repeat Required
9ab6c0d9	fail	92.82	64.8	no	no	yes
e5ffea1d	fail	92.9	264.4	yes	no	yes
0ab71d84	fail	93.93	173.4	no	no	yes
1a574171	fail	94.17	197.5	no	no	yes
716130a0	fail	93.56	151.6	no	no	yes
078d9814	not_reported	n/a	n/a	no	no	yes
b36fc4ca	fail	94.61	89.5	no	no	yes
ce1cd7ce	fail	92.23	275.3	no	no	yes
740131f2	not_reported	n/a	n/a	no	no	yes
e0b496a8	not_reported	n/a	n/a	no	no	yes
24cc59c8	fail	92.66	73.4	no	no	yes
89f900c2	not_reported	n/a	n/a	no	no	yes

Table 15. QC drill-down identifying samples requiring repeat testing or cautious interpretation before operational decisions.

Run-Level QC Summary

Sequencing run quality directly affects confidence in transmission inferences. Runs with high failure rates or low mean depth should trigger laboratory review before operational decisions are made from that run's samples. NOTE: run identifiers are not recorded in this extract; rows below are grouped by the report date of associated QC records as a run proxy.

Table 16. Run-level QC summary aggregated by report date. Runs with fail rate >20% or mean coverage <95% should be reviewed before results are used for cluster assignment. Assign run identifiers to sample_qc_metrics.run_id to enable full run-level audit.

Drug-Resistance Mutation Details

WARNING: The mapping between predicted mutations and drug resistance is preliminary. All genomic resistance predictions must be confirmed by phenotypic DST (drug susceptibility testing) before clinical use. Invalid gene-drug combinations are flagged below and should not be reported operationally until pipeline validation is complete.

Drug	Expected genes	Common unusual/invalid pairings to review
Rifampicin	rpoB	gyrA, pncA, katG, inhA — flag for pipeline review
Isoniazid	katG, inhA, fabG1	gyrA, rpoB — flag for pipeline review
Pyrazinamide	pncA	gyrA, katG, rpoB, embB — flag for pipeline review
Ethambutol	embB	pncA, katG, rpoB — flag for pipeline review
Fluoroquinolones	gyrA, gyrB	inhA, embB, katG — flag for pipeline review
Aminoglycosides / injectables	rrs, eis, tlyA	rpoB, pncA — flag for pipeline review

Table 17. Minimum expected gene-drug mapping reference (WHO/standard TB resistance catalogue). Pairings outside the Expected genes column indicate pipeline misconfiguration and must be resolved before clinical reporting. Validate all calls against WHO TB mutation catalogue v2 or equivalent.

Case	Drug	Mutation	Gene	Gene-drug status	Confidence	Predicted	DST
0ab71d84	pyrazinamide	S531L	gyrA	Unusual gene-drug mapping	n/a	pyrazinamide	not_available
24cc59c8	rifampicin	H526Y	gyrA	Unusual gene-drug mapping	n/a	rifampicin	not_available
716130a0	pyrazinamide	Q431K	katG	Unusual gene-drug mapping	n/a	pyrazinamide, fluoroquinolones	not_available
716130a0	fluoroquinolones	C15T	inhA	Unusual gene-drug mapping	n/a	pyrazinamide, fluoroquinolones	not_available
78e6e3e1	rifampicin	C15T	pncA	Unusual gene-drug mapping	n/a	rifampicin	not_available
9ab6c0d9	pyrazinamide	D516V	gyrA	Unusual gene-drug mapping	n/a	pyrazinamide	not_available
a8d08022	ethambutol	S315T	pncA	Unusual gene-drug mapping	n/a	ethambutol	not_available
b36fc4ca	pyrazinamide	C15T	gyrA	Unusual gene-drug mapping	n/a	pyrazinamide, fluoroquinolones	not_available
b36fc4ca	fluoroquinolones	S531L	embB	Unusual gene-drug mapping	n/a	pyrazinamide, fluoroquinolones	not_available
b6530faa	pyrazinamide	Q431K	rpoB	Unusual gene-drug mapping	n/a	pyrazinamide	not_available
e0b496a8	fluoroquinolones	H526Y	gyrA	Valid expected gene	n/a	fluoroquinolones	not_available
e10f9dde	isoniazid	H526Y	inhA	Valid expected gene	n/a	isoniazid	not_available
e5ffea1d	isoniazid	S531L	gyrA	Unusual gene-drug mapping	n/a	isoniazid	not_available

Table 18. Drug-resistance mutation evidence. Gene-drug status is classified as Valid expected gene, Unusual gene-drug mapping, or Unknown / catalogue not available. WARNING: Do not report invalid gene-drug pairs as clinical resistance until the pipeline mapping is corrected. All resistance predictions require phenotypic DST confirmation before operational action.

Phenotypic DST Reconciliation

Genomic resistance predictions must be reconciled against phenotypic drug susceptibility testing (DST) before any clinical or operational decision is made. Discordance (e.g. genomic resistance predicted but phenotypic DST sensitive) requires immediate laboratory and clinical review. **Phenotypic DST data is NOT recorded in this system extract.** Until phenotypic results are integrated, all genomic resistance signals must be treated as preliminary flags only. Populate the cases or tb_interpretation table with phenotypic DST results to enable automated reconciliation.

Table 19. Phenotypic DST reconciliation table. Genomic predictions are derived from TBProfiler pipeline output. Phenotypic DST column will be populated when lab results are integrated. Until phenotypic results are available, do not report genomic resistance as confirmed susceptibility or resistance for prescribing decisions.

Cluster Epidemiology Summary

The cluster tracker is presented in two tables: a genomic summary (dates, SNP distances, resistance burden) and an operational tracker (investigation lead, epi link, contact tracing, LTBI, DST, next step).

Cluster	Cases	First	Latest	Med SNP	Max SNP	RR/MDR	Index case	Recent 30/60/90d
04e65e42	6	2024-11-28	2026-05-02	25.0	25	2/2	3a23165b	1/1/3
39be6563	6	2025-03-20	2026-01-25	25.0	25	3/3	fb419fc3	0/0/0
604b7f43	2	2025-01-29	2026-01-05	25.0	25	1/1	b6530faa	0/0/0

Table 20. Cluster genomic summary: case counts, specimen date range, SNP distance range, RR/MDR case burden, index case, and recent case counts (30/60/90 days).

Cluster	Status	Lead	Epi link	Setting	Contact tracing	LTBI screen	DST status	Next step
04e65e42	open+linked	Pending	Pending	Unknown	Not started	No	Pending	Review at MDT
39be6563	open+linked	Pending	Pending	Unknown	Not started	No	Pending	Review at MDT
604b7f43	open+linked	Pending	Pending	Unknown	Not started	No	Pending	Review at MDT

Table 21. Cluster operational tracker: investigation lead, epi link status, setting, contact tracing, LTBI screening, DST status, recommended next step. Fields pre-populated as 'Pending' must be completed by the responsible clinician or investigator before MDT circulation.

Cluster Growth Status

Cluster growth status classifies whether each cluster is actively accumulating new cases based on recent specimen dates. Active clusters (last case <90 days ago) require heightened investigation priority. Clusters with no new cases for >180 days may be approaching natural closure but require formal MDT sign-off.

Table 22. Cluster growth status. +30d/+60d/+90d = cases with specimen date within that window of today. Active = last case <90 days ago; Slowing = 90–180 days; Likely inactive = >180 days. Status does not imply transmission has stopped; formal closure requires MDT sign-off against cluster closure criteria.

Epi-Link Evidence Summary

Genomic cluster membership is necessary but not sufficient for establishing a transmission chain. Epidemiological links — shared contacts, common exposure settings, overlapping timelines — provide independent corroboration. The table below summarises epi-link status for all model-prioritised pairs from this run's output.

Epi-link category	Pair count	% of all pairs
Same Cluster	10	55.6%
Not Confirmed	8	44.4%

Epi-link corroboration rate: 18 of 18 pairs (100.0%) have any recorded epi link. Pairs without epi corroboration require field investigation before operational escalation.

Contact-Tracing Yield

Contact-tracing yield measures the productivity of outbreak investigation activities: how many contacts were identified per index case, how many were tested, and how many were positive. **Contact-tracing data is NOT recorded in this system extract.** The table below shows clusters with placeholder values. Field investigation teams should populate this data via the case management system and re-ingest before MDT circulation.

Table 23. Contact-tracing yield by cluster. Contact tracing data must be provided by field investigation teams and integrated before operational use. Yield % = (test positive / contacts tested) × 100. All values 'Not recorded' until data are supplied by the field team.

Geographical Cluster Spread

Cluster	Regions	Cases	Cross-Region	Risk Flag
04e65e42	Regional geography unavailable in this extract	6	no	high
39be6563	Regional geography unavailable in this extract	6	no	high
604b7f43	Regional geography unavailable in this extract	2	no	high

Table 24. Cluster geography summary at region/trust-safe level (no household-level identifiers displayed).

Regional geography unavailable in this extract; data is recorded at United Kingdom level only. For operational surveillance, re-ingest with a safe hierarchy in geographic_region: HSC Trust, PHA locality, council area, and postcode district only if governance approvals and small-number controls are satisfied.

Missing-Data Dashboard

Field	Completeness	Missing	Impact
Collection date	100.0%	0	affects outbreaker2 timing
Notification date (proxy)	100.0%	0	affects surveillance timeline
Region/trust	100.0%	0	affects representativeness and spread
Lineage	100.0%	0	affects phylogenetic interpretation
Resistance call	100.0%	0	affects clinical action
QC metrics	83.3%	4	affects confidence in inference

Table 25. Completeness and impact dashboard for key outbreak-investigation data fields, aligned to transparent inference limitations.

Cluster Closure Criteria

Clusters should not be closed on genomic evidence alone. The following criteria define the minimum standards for cluster closure, based on ECDC/PHE TB cluster management guidance. All criteria should be met before a cluster is formally closed at MDT. Closure is documented with MDT sign-off and rationale.

Criterion	Standard	How to assess	Met in this report?
No new linked cases	No genomically linked new case within preceding 12 months	Review cluster growth status table; confirm last case >365 days ago	Review cluster growth status table
All contacts traced	All known contacts identified, assessed, and outcome recorded	Contact-tracing yield table shows 100% contacts assessed	Contact tracing data not recorded in this extract
Index case identified or excluded	Probable index case or documented exclusion of identifiable source	Transmission network and MDT review	Probable index case may be noted in cluster operational tracker
Transmission links resolved	All SNP-linked pairs adjudicated (epi confirmed or excluded)	Epi-link evidence summary: all pairs have documented epi outcome	Epi-link data partially available — see epi-link summary section
Drug resistance resolved	Phenotypic DST completed for all resistance-predicted cases	Phenotypic DST reconciliation table: no 'Awaiting' entries	Phenotypic DST not recorded in this extract
MDT sign-off documented	Formal MDT discussion with closure rationale recorded	Case management system or minutes reference	Out of scope for automated report

Cluster closure must be documented in the case management system with date, responsible clinician, and rationale. Genomic surveillance should continue after closure to detect re-emergence.

Model Reliability Diagnostics

MCMC convergence diagnostics indicate whether the Bayesian sampler has explored the parameter space adequately. Convergence diagnostic (Gelman-Rubin / R-hat) <1.1 indicates reliable estimates. Values >1.1 suggest exploratory inference only. Effective sample size, acceptance rate, and multi-chain robustness all support confidence in the transmission probabilities reported. For robust surveillance use: fixed random seed, multiple chains, longer run length, reported R-hat/ESS/acceptance rate, and sensitivity analysis using plausible TB generation-time priors.

Diagnostic	Value	Status	Meaning
Convergence (R-hat)	n/a	■ Review	Not available
Effective sample size (ESS)	n/a	■ Limited	Not available
Acceptance rate	n/a	■ Check	Not available
Parallel chains	n/a	■ Single-chain	Single-chain: less robust
Sensitivity analysis	Not Performed	■ No	Confirms results stable across parameter priors

Table 26. MCMC convergence diagnostics. Use exploratory language for transmission interpretation if convergence R-hat >1.1. All reported posterior probabilities should be interpreted with awareness of these diagnostic results.

Transmission Routes Reference

Understanding TB Transmission Routes from WGS

Whole-genome sequencing identifies genomic relatedness but does not directly observe contact events. Combining genomic clusters with epidemiological data (contact tracing, shared locations, timeline of diagnosis) allows investigators to build a plausible transmission chain. The table below summarises the genomic signals and their transmission implications.

Genomic Signal	SNP Range	Transmission Implication	Recommended Action
Highly probable direct transmission	0–5 SNPs	Strong genomic evidence of recent direct person-to-person transmission. Strain has had little time to evolve.	Immediate contact tracing; identify shared setting (household, workplace, healthcare).
Possible direct or near-direct transmission	6–12 SNPs	Genetically close; consistent with transmission within the last 1–3 years or via an undetected intermediate.	Epidemiological linkage investigation; check whether cases share contacts or settings.
Within extended cluster — indirect link likely	13–50 SNPs	Genetically related but too diverged for recent direct transmission. Likely share a common ancestor strain.	Review cluster history; may represent reactivation from the same source years earlier.
Unrelated strains	>50 SNPs	No plausible genomic link. Coincident diagnoses are likely due to independent exposure or reactivation.	No cluster-based action; manage as separate cases.
Mixed-lineage / contamination	N/A	Two or more distinct strain signals in a single sample. May indicate laboratory cross-contamination or mixed infection.	Flag for repeat sequencing; do not use in cluster assignments until resolved.

Table 27. TB transmission route classification by SNP distance (M. tuberculosis whole-genome comparison). SNP thresholds follow UK NICE guideline NG33 and published literature (Walker et al. 2013, Meehan et al. 2019). The generation-time prior used in outbreaker2 is programme-configured and influences whether a given SNP distance is interpreted as consistent with direct versus indirect transmission.

Transmission Priority Signals

Case	Region	Risk	Out	In
e10f9dde		2.316	3	1
3a23165b		1.8442	3	1
e0b496a8		1.6958	2	1
fb419fc3		1.6042	2	1
a8d08022		1.603	2	1
89f900c2		1.4	2	0
ce1cd7ce		1.0803	2	1
24cc59c8		1	1	1
78e6e3e1		1	1	1
f17d7f92		1	1	1

Table 28. Top-priority cases by network centrality. Network snapshot: 24 nodes, 23 directed links, 18 model-prioritised links (posterior >0.70). **Out** = outgoing transmission links (potential sources); **In** = incoming links (potential recipients). Cases with multiple outgoing model-prioritised links should be treated as potential source nodes requiring validation, not confirmed sources. Counts may differ where links are filtered for display, QC status, or posterior thresholding. Executive summary count reflects all model links (18); network snapshot JSON count is 18.

Model-prioritised links with posterior probability >0.70 represent statistical transmission hypotheses from outbreaker2. In this run, they should not be treated as genomic evidence of direct transmission unless supported by pairwise SNP distance, QC pass status, and epidemiological corroboration.

Visual element	Operational meaning
Node colour	Cluster assignment (same colour = same cluster context)
Red border	QC unresolved (failed, contaminated, or not reported)
RR/MDR/FQ marker	Predicted resistance signal; confirm by phenotypic DST
Solid edge	Pairwise SNP ≤12 and QC pass (supported candidate link)
Dotted edge	Pairwise SNP >12 (unlikely direct recent transmission)
Grey edge	Pairwise SNP unavailable (model-only hypothesis)

Table 29. Recommended visual encoding for future network outputs. Note: not all elements below may be visible in the current network figure depending on pipeline configuration. This encoding should be applied when network graphs are regenerated with full SNP and QC data.

Diagnostic Graphics

The following plots are generated by outbreaker2 and the platform's supplementary visualisation pipeline. Each figure caption explains the content and how to interpret the output.

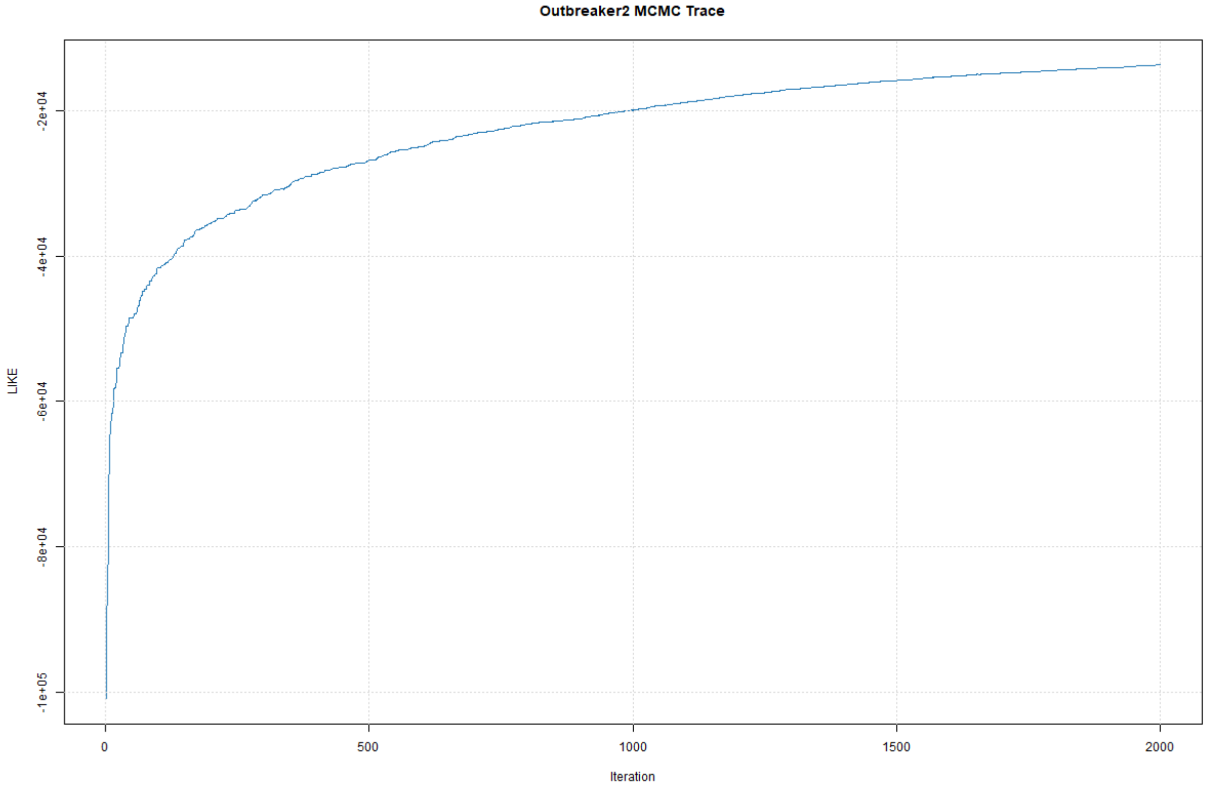


Figure 2. MCMC trace plot — log-posterior probability over iterations. In this run, the trace does not clearly demonstrate stable mixing; outbreaker2-derived directionality should therefore be treated as exploratory pending repeat runs with longer chains and formal convergence diagnostics.

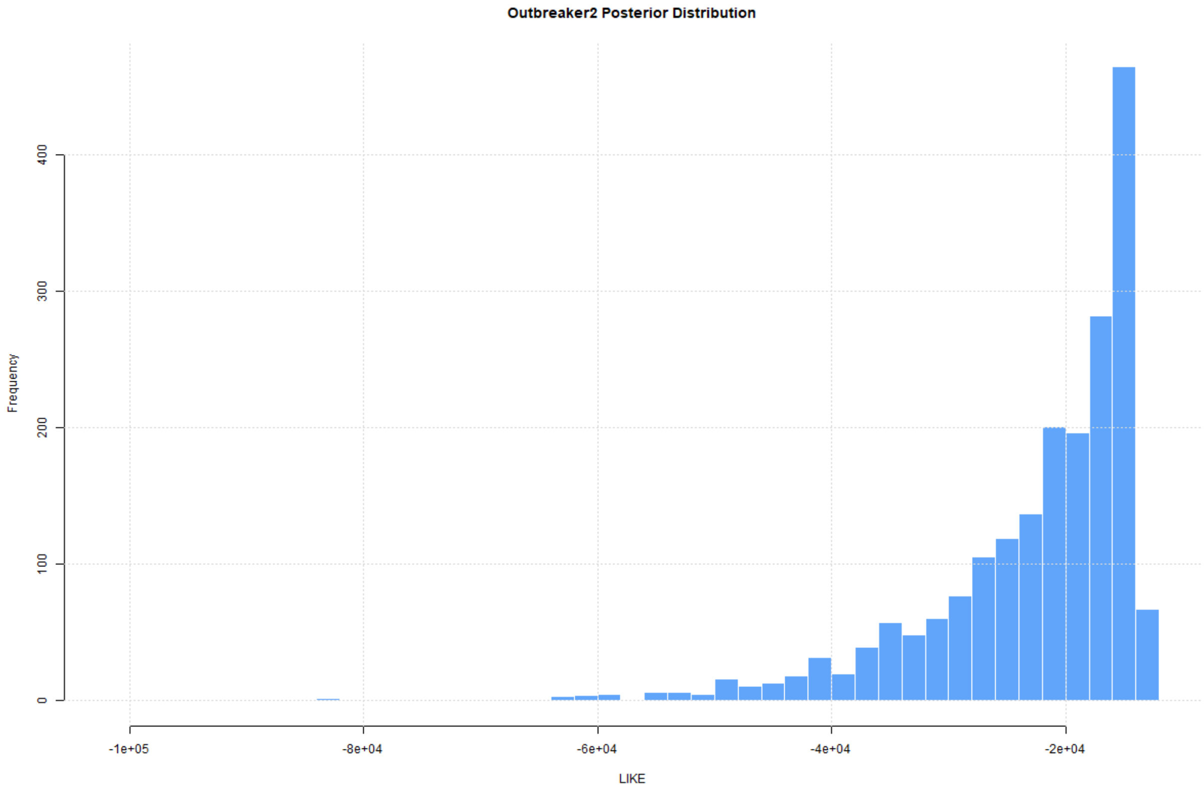
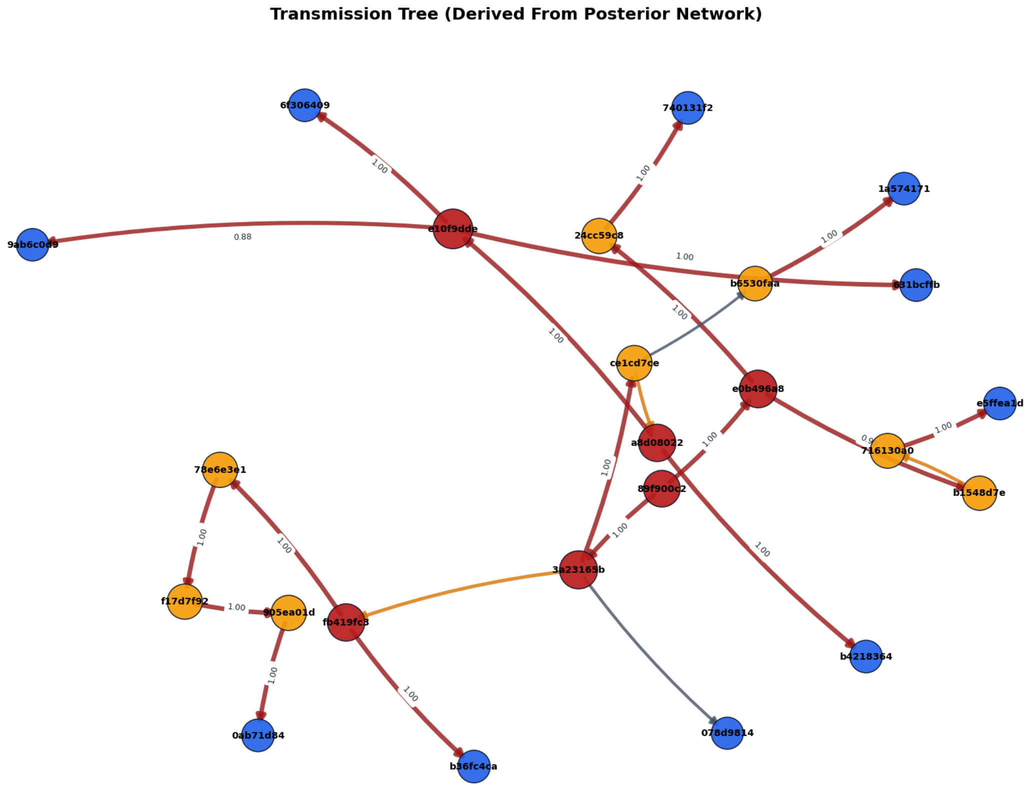
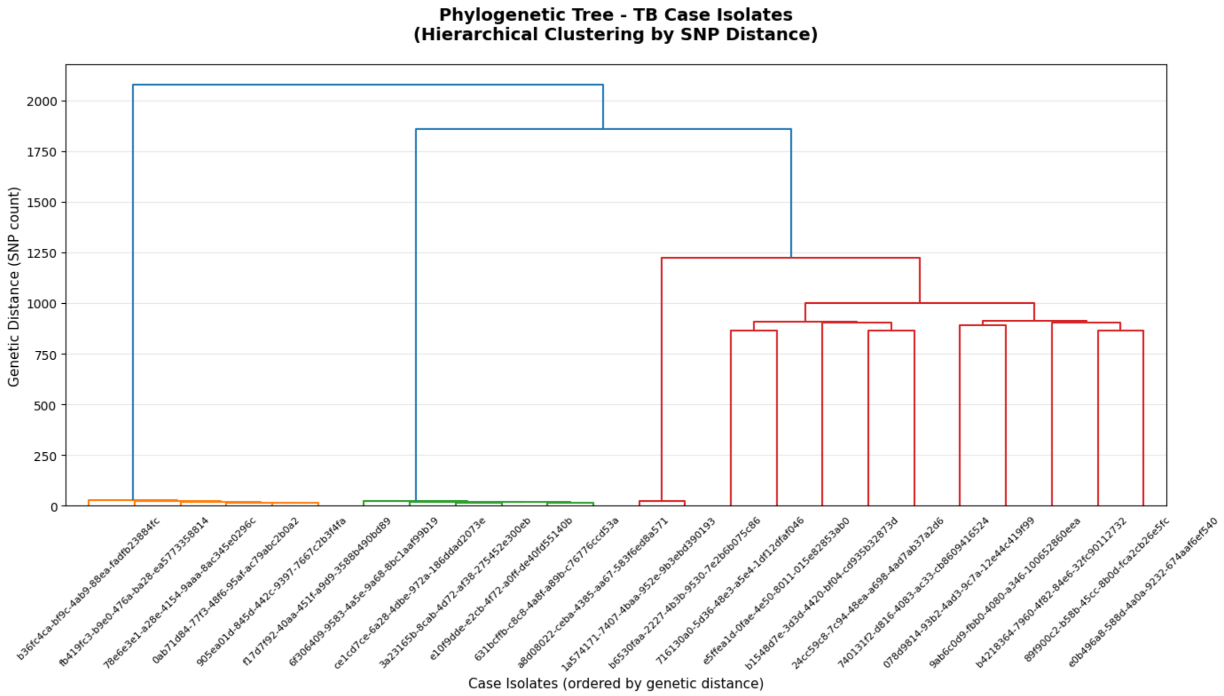


Figure 3. Posterior distribution histograms — marginal distributions of key model parameters (transmission probability, sampling probability, generation time). Narrow, symmetric peaks indicate well-determined parameters. Broad or multi-modal distributions suggest parameter uncertainty, which should be reflected in cautious interpretation of individual transmission links.



Embedded legend	Encoding
Node colour	Cluster
Red border	QC unresolved
Dotted edge	SNP >12
Grey edge	SNP unavailable
Solid edge	SNP ≤12 and QC pass
RR/MDR marker	Predicted resistance; confirm with DST

Figure 4. Inferred transmission tree (most probable who-infected-whom). Each node is a case; arrows indicate the direction of inferred transmission from source (tail) to recipient (head). Arrow thickness or colour (where shown) reflects posterior probability. Dashed or thin arrows indicate lower-confidence links. Cases with no incoming arrow are probable index cases or represent undetected importation events.



Embedded legend	Encoding
Node colour	Cluster
Red border	QC unresolved
Dotted edge	SNP >12
Grey edge	SNP unavailable
Solid edge	SNP ≤12 and QC pass
RR/MDR marker	Predicted resistance; confirm with DST

Figure 5. Phylogenetic context — a midpoint-rooted maximum parsimony or neighbour-joining tree of sequenced cases, coloured by cluster or region. Branch length represents SNP distance. Cases on short branches with few SNPs between them form tight clades consistent with recent transmission. Well-separated clades indicate genetically distinct strain lineages circulating concurrently.

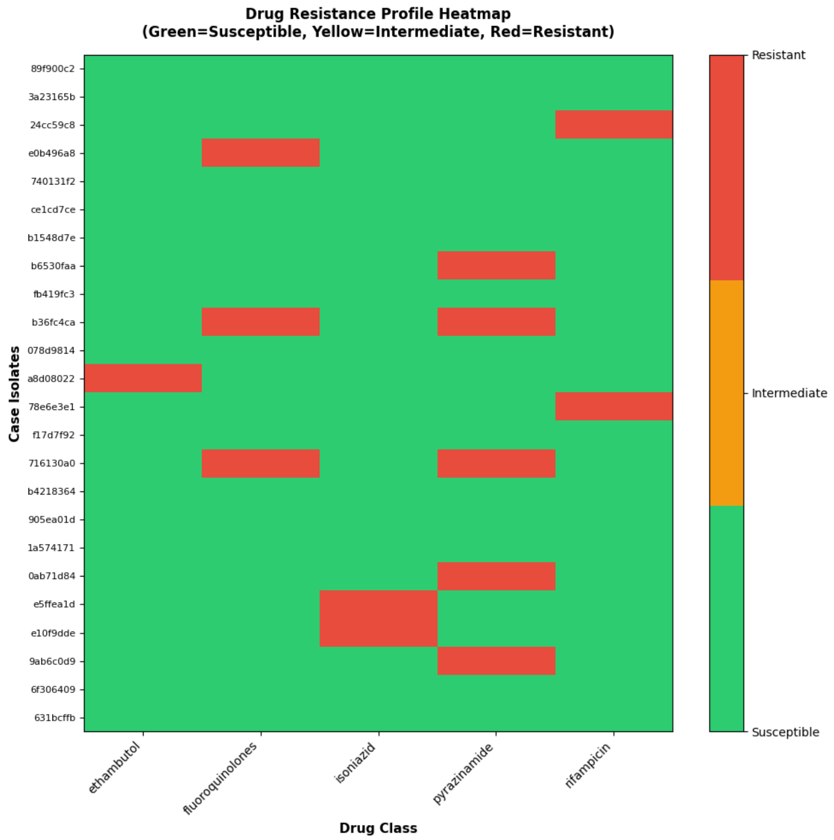


Figure 6. Drug resistance profile summary — frequency of predicted resistance mutations across the sequenced cohort. Bars represent the proportion of cases with predicted resistance to each antibiotic class. Rifampicin + isoniazid co-resistance defines MDR-TB. High frequencies of any first-line resistance warrant urgent review of empirical treatment protocols.

M. tuberculosis Lineage Reference

Lineage classification places a strain within the global phylogeny of M. tuberculosis. The table below provides context for lineages commonly observed in Northern Ireland.

Lineage	Name / Origin	DR Association	NI Relevance	Notes
L1	East-African-Indian / Indo-Oceanic	Lower MDR-TB frequency	Cases linked to South Asia, Horn of Africa	Commonly found in Bangladeshi, Indian, Somali communities.
L2	East-Asian (Beijing lineage)	High MDR/XDR-TB risk; associated with resistance acquisition	Sporadic importation; watch for resistance	Beijing strains have shown high transmissibility in some outbreak settings.
L3	East-African-Indian (Delhi/CAS)	Moderate DR frequency	Cases linked to Pakistan, Afghanistan, India	Common in large urban TB programmes in the UK.
L4	Euro-American	Generally lower DR; but historical MDR clusters exist	Dominant UK-born strain type	Most legacy UK TB is L4. Reactivation common in older cohorts.
L5 / L6	West-African (Mycobacterium africanum)	Lower overall DR	Cases linked to West Africa	Slower growth, may present with atypical features.
L7	Ethiopian	Limited data	Rare in NI	Emerging lineage classification; limited clinical guidance available.

Table 30. M. tuberculosis lineage reference. DR = drug resistance; MDR = multidrug-resistant; XDR = extensively drug-resistant. Lineage assignment should be combined with phenotypic DST and clinical judgment. Source: Coll et al. (2014) Nature Genetics; WHO Global TB Report 2023.

Clinical and Public Health Action Summary

The table below maps genomic findings to recommended clinical and public health actions. All actions must be confirmed by the responsible clinician and public health team.

Finding	Recommended Action	Urgency
New case links to an existing open cluster (≤12 SNPs)	Notify cluster lead; extend contact tracing to include new case contacts; review whether the cluster source has been identified.	Within 5 working days
Model-prioritised link identified (posterior >0.70)	Review only after confirming pairwise SNP support (≤12), QC pass status, and epidemiological plausibility. Do not escalate on model probability alone; document whether SNP and epi evidence corroborates the link.	Within 5 working days
New cluster opened (≥2 cases genetically linked, no prior cluster)	Open investigation; notify public health; assign epidemiologist; initiate contact tracing for all cases.	Within 2 working days
MDR-TB predicted by genomics	Confirm with phenotypic DST; notify MDR-TB specialist centre; initiate enhanced infection control if hospitalised.	Immediately on result
Sequencing coverage <80% for a region	Review specimen submission and transport processes for that region; identify cases that did not receive WGS and arrange retrospective sequencing if available.	Monthly programme review
MCMC convergence diagnostic >1.1	Do not rely on posterior transmission probabilities from this run. Increase MCMC iterations and re-run outbreaker2. Check input data completeness.	Before using results
Contamination flag on a sample	Exclude sample from cluster assignments; arrange repeat sequencing from original culture. Investigate laboratory process if multiple consecutive contamination flags.	Within 10 working days

Table 31. Recommended clinical and public health actions mapped to genomic findings. Urgency thresholds align with PHE/PHA TB operational guidance. All genomic findings must be reviewed in conjunction with clinical history, contact tracing records, and microbiological DST before action is taken.

Disclaimer: Genomic cluster assignments and transmission inferences are probabilistic estimates based on mathematical models. They supplement but do not replace epidemiological investigation. Do not use genomic evidence alone to assign legal or clinical liability for transmission.

Data Provenance

This report combines outbreaker outputs with surveillance KPIs, lineage/DR validation, secondary engine readiness, and cross-method clustering comparison artifacts available at generation time.
Outbreaker summary timestamp: 2026-05-11T15:10:31Z
Transmission network timestamp: 2026-05-11T15:10:31Z
Lineage/DR validation timestamp: 2026-05-11T15:09:18.782756+00:00
Method comparison timestamp: 2026-05-11T15:10:36.749114+00:00

Metadata item	Value
Reference genome	Not recorded in this extract — mandatory before external circulation (expected: H37Rv / NC_000962.3)
SNP-calling pipeline/version	Not recorded in this extract — mandatory before external circulation (e.g. Clockwork/COMPASS/Snippy + version)
Resistance catalogue/version	Not recorded in this extract — mandatory before external circulation (e.g. WHO mutation catalogue v2)
Lineage-calling tool/version	Not recorded in this extract — mandatory before external circulation (e.g. TB-Profiler v4.x / Mykrobe)
outbreaker2 version	Not recorded — engine: outbreaker2 (mandatory before external circulation)
Random seed	Not set — run is non-reproducible without a fixed seed; mandatory before external circulation
Model priors	Default outbreaker2 priors; MCMC: 2000 generations, burnin 500, 1500 posterior samples
Input hash: exports/cases.csv	159441304aaff8e0cb519faec6d1788aa43a142668cc49f7cffa1c55f4530137
Input hash: exports/dna.fasta	c490a2b8db4cd6e86213a1b5c27e6c8c718562ac81fb38a99429eb7df4e6a7b1
Input hash: exports/transmission_network.json	b7b0e7f6b5bbfc4aa487246f06dca88ce769cc1cf343a59f2a322366e81523ec

Table 32. Software and reproducibility metadata for governance and audit traceability.

PRE-CIRCULATION GOVERNANCE NOTICE: 6 reproducibility field(s) are unpopulated: Reference genome; SNP-calling pipeline/version; Resistance catalogue/version; Lineage-calling tool/version; outbreaker2 version; Random seed. External/formal circulation is blocked until these are completed. Random seed must be recorded to ensure reproducibility. Contact the pipeline administrator before proceeding.

Companion files: exports/appendix_a_case_level_actions.csv — full case-level actions; exports/appendix_b_full_discordance_review.csv — full discordance table; exports/transmission_pairs.csv — all model-prioritised pairs with owner/due fields.

Appendix A: Case-Level Operational Action Details

Table A1a — classification (cluster, SNP, QC, tier). Table A1b — actions (likely link, warning code, action code). Full 10-column data: exports/appendix_a_case_level_actions.csv.

Table A1a — Case Classification

Case	Cluster	Pairwise SNP?	NN SNP	Posterior	Tier	QC
631bcffb	04e65e42	yes	15	1.000	Moderate confidence	pass
6f306409	04e65e42	yes	17	1.000	Moderate confidence	pass
9ab6c0d9	none	yes	883	0.880	Exploratory	fail
e10f9dde	04e65e42	yes	13	1.000	Moderate confidence	pass
e5ffea1d	none	yes	864	1.000	Exploratory	fail +contam
0ab71d84	39be6563	yes	15	1.000	Exploratory	fail
1a574171	604b7f43	yes	22	1.000	Exploratory	fail
905ea01d	39be6563	yes	13	1.000	Moderate confidence	pass
b4218364	none	yes	879	1.000	Exploratory	pass
716130a0	none	yes	864	1.000	Exploratory	fail
f17d7f92	39be6563	yes	13	1.000	Moderate confidence	pass
78e6e3e1	39be6563	yes	15	1.000	Moderate confidence	pass
a8d08022	04e65e42	yes	16	1.000	Exploratory	pass
078d9814	none	no	n/a	0.525	Exploratory	not_reported
b36fc4ca	39be6563	yes	22	1.000	Exploratory	fail
fb419fc3	39be6563	yes	17	1.000	Moderate confidence	pass
b6530faa	604b7f43	yes	22	1.000	Moderate confidence	pass
b1548d7e	none	yes	884	0.994	Exploratory	pass
ce1cd7ce	04e65e42	yes	16	1.000	Exploratory	fail
740131f2	none	no	n/a	1.000	Exploratory	not_reported
e0b496a8	none	no	n/a	1.000	Exploratory	not_reported
24cc59c8	none	yes	884	1.000	Exploratory	fail
3a23165b	04e65e42	yes	13	1.000	Exploratory	pass
89f900c2	none	no	n/a	1.000	Exploratory	not_reported

Table A1a. Case classification: cluster assignment, pairwise SNP availability, nearest-neighbour SNP, outbreaker2 posterior, confidence tier, and QC status.

Table A1b — Case Actions

Warning codes: SNP-missing/QC — QC unresolved, pairwise SNP unavailable; Model-only — outbreaker2 link, no pairwise SNP; SNP-linked — SNP ≤12, epi corroboration still required; SNP>12 — pairwise SNP above transmission threshold. **Action codes:** Repeat/QC — repeat or verify sequence before any transmission interpretation; Validate SNP+epi — confirm with pairwise SNP and epidemiology before operational action. Full text is in exports/appendix_a_case_level_actions.csv.

Table A1b. Case actions (coded).

Case	Likely link	Warning	Recommended action
631bcffb	e10f9dde -> 631bcffb	SNP>12	Validate SNP+epi
6f306409	e10f9dde -> 6f306409	SNP>12	Validate SNP+epi
9ab6c0d9	e10f9dde -> 9ab6c0d9	SNP-missing/QC	Repeat/QC
e10f9dde	a8d08022 -> e10f9dde	SNP>12	Validate SNP+epi
e5ffea1d	716130a0 -> e5ffea1d	SNP-missing/QC	Repeat/QC
0ab71d84	905ea01d -> 0ab71d84	SNP-missing/QC	Repeat/QC
1a574171	b6530faa -> 1a574171	SNP-missing/QC	Repeat/QC
905ea01d	f17d7f92 -> 905ea01d	SNP>12	Validate SNP+epi
b4218364	a8d08022 -> b4218364	SNP>12	Validate SNP+epi
716130a0	716130a0 -> e5ffea1d	SNP-missing/QC	Repeat/QC
f17d7f92	78e6e3e1 -> f17d7f92	SNP>12	Validate SNP+epi
78e6e3e1	fb419fc3 -> 78e6e3e1	SNP>12	Validate SNP+epi
a8d08022	a8d08022 -> b4218364	SNP>12	Validate SNP+epi
078d9814	3a23165b -> 078d9814	SNP-missing/QC	Repeat/QC
b36fc4ca	fb419fc3 -> b36fc4ca	SNP-missing/QC	Repeat/QC
fb419fc3	fb419fc3 -> b36fc4ca	SNP>12	Validate SNP+epi
b6530faa	b6530faa -> 1a574171	SNP>12	Validate SNP+epi
b1548d7e	e0b496a8 -> b1548d7e	Model-only	Validate SNP+epi
ce1cd7ce	3a23165b -> ce1cd7ce	SNP-missing/QC	Repeat/QC
740131f2	24cc59c8 -> 740131f2	SNP-missing/QC	Repeat/QC
e0b496a8	89f900c2 -> e0b496a8	SNP-missing/QC	Repeat/QC
24cc59c8	e0b496a8 -> 24cc59c8	SNP-missing/QC	Repeat/QC
3a23165b	89f900c2 -> 3a23165b	Model-only	Validate SNP+epi
89f900c2	89f900c2 -> 3a23165b	SNP-missing/QC	Repeat/QC

Appendix B: Full Discordance Review

Discordance codes classify pairs where pairwise SNP evidence and outbreaker2 linkage disagree. Full data including verbose interpretation is in exports/appendix_b_full_discordance_review.csv.

Discordance Code Definitions

Code	Meaning
D1	Model-linked (outbreaker2 ≥0.70) AND pairwise SNP >12 — genomically discordant; unlikely direct recent transmission. Do not escalate without further review.
D2	Model-linked (outbreaker2 ≥0.70) AND pairwise SNP unavailable — requires sequencing/SNP analysis before transmission interpretation.
D3	Pairwise SNP ≤12 but NOT model-prioritised — possible older or shared-source linkage; review epidemiology to determine significance.

Case Pair	Pairwise SNP	Posterior	Code
0ab71d84-905ea01d	15	1.000	D1
3a23165b-89f900c2	n/a	1.000	D2
716130a0-e5ffea1d	864	1.000	D1
1a574171-b6530faa	22	1.000	D1
a8d08022-e10f9dde	17	1.000	D1
631bcffb-e10f9dde	16	1.000	D1
24cc59c8-740131f2	n/a	1.000	D2
78e6e3e1-f17d7f92	16	1.000	D1
6f306409-e10f9dde	18	1.000	D1
3a23165b-ce1cd7ce	16	1.000	D1
b36fc4ca-fb419fc3	28	1.000	D1
78e6e3e1-fb419fc3	21	1.000	D1
a8d08022-b4218364	881	1.000	D1
24cc59c8-e0b496a8	n/a	1.000	D2
905ea01d-f17d7f92	13	1.000	D1
89f900c2-e0b496a8	n/a	1.000	D2
b1548d7e-e0b496a8	n/a	0.994	D2
9ab6c0d9-e10f9dde	883	0.880	D1
3a23165b-fb419fc3	889	0.681	D1
a8d08022-ce1cd7ce	19	0.677	D1
716130a0-b1548d7e	884	0.661	D1
078d9814-3a23165b	n/a	0.525	D2
b6530faa-ce1cd7ce	898	0.438	D1

Table B1. Discordant pairs (coded). Full data including verbose interpretation in exports/appendix_b_full_discordance_review.csv.

Appendix C: One-Page MDT Action Sheet

Priority area	Current signal	Required MDT action	Owner	When
Circulation readiness	Development / Internal Draft Only	Complete mandatory reproducibility metadata before external circulation	Pipeline + Governance	Before circulation
Model reliability	Exploratory	Treat directionality as exploratory until diagnostics are complete	Bioinformatics + MDT	Current cycle
Open clusters: 3; high-priority (>10): 2	3 total / 2 >10	Ensure MDT review and epidemiological data completion for all open clusters. Do not infer source-recipient direction from model output alone.	MDT + Field team	Next MDT
Discordant model links	23	Use pairwise SNP + epidemiology adjudication pathway	MDT	Next MDT
Geography completeness	UK-only extract	Populate HSC Trust, PHA locality, council area; use postcode district only with governance approval	Data management	Next ingest

Table C1. Condensed MDT action sheet for governance and operational review.

Appendix D: TB Genomics Key Concepts

Reference definitions for genomic terminology used throughout this report.

Concept	Explanation
Whole-Genome Sequencing (WGS)	Reads the complete ~4.4 Mb genome of M. tuberculosis. More informative than conventional typing (MIRU, spoligotyping).
SNP (single nucleotide polymorphism)	A single base-pair difference in the genome. Closely related strains share very few SNPs. Used as a genetic 'distance' metric.
SNP threshold for transmission	Strains with ≤12 SNPs are considered potentially linked (UK NICE guidance). ≤5 SNPs suggests recent direct transmission. >50 SNPs effectively rules out recent shared transmission.
Lineage	M. tuberculosis is classified into 7+ major lineages (L1–L7) reflecting global evolutionary history. Lineage influences drug-resistance patterns and may correlate with transmissibility.
Cluster	A group of cases whose sequences are genetically similar (within the SNP threshold). A cluster does not prove direct person-to-person transmission — epidemiological linkage is required to confirm transmission routes.
outbreaker2	A Bayesian MCMC method that combines SNP distances with collection dates and an assumed generation time to probabilistically infer who-infected-whom. Output posterior probabilities indicate the likelihood of a direct transmission event between any pair of cases.
Generation time	The average time between one person being infected and the next person they infect being detected. For TB, this is typically 1–3 years (range 0.5–5 years) due to the long latency period.
MCMC convergence	Markov Chain Monte Carlo simulations must reach a stable state ('converge'). A convergence diagnostic near 1.0 indicates reliable estimates. Values >1.1 suggest the chain has not fully mixed and results should be interpreted cautiously.
Drug resistance	Genomic mutations predict resistance to first-line drugs (isoniazid, rifampicin, etc.) and define MDR-TB (multi-drug resistant) and XDR-TB (extensively drug resistant). Genomic DR prediction is used alongside phenotypic DST.
Transmission network	A directed graph where arrows indicate the most probable direction of transmission. Model-prioritised links (posterior probability >0.70) are hypotheses and should be validated against pairwise SNP, QC status, and epidemiology.

Table D1. TB genomics reference — key terms (WHO/UK TB genomic surveillance guidance).